

A cut-cell immersed boundary for *moving* walls

Mass conservation, the discrete geometric conservation law, and what it costs

September 8, 2026

Abstract

We extend the cut-cell immersed-boundary method of `doc/freeslip_ibm_part2.pdf` to a rigid body that moves through a fixed Cartesian mesh. The mesh never moves, so the method is not an ALE scheme: no face carries a mesh velocity, and the only trace of the motion is what the wall segment of a cut cell does. Applying Reynolds transport to the fluid part of such a cell shows that *continuity is the only equation that changes*: it acquires a source equal to the flux the wall sweeps. Every other term keeps its stationary form, because the wall is impermeable in its own frame. Free slip on a moving wall then needs no new machinery at all – the source *is* the boundary condition – and no slip needs one explicit term. The central question is which of two equivalent-looking quantities is primary: the geometric volume change $(\alpha^{n+1} - \alpha^n)V/\Delta t$, or the discrete wall flux $-\mathbf{u}_b \cdot \mathbf{S}_w$. We show that only the second works, that taking the volume change from it rather than the other way round makes the discrete geometric conservation law hold by construction, and that the resulting scheme preserves a free stream to machine precision at any body speed and from the first outer iteration, conserves mass to roundoff with no compatibility projection anywhere, and requires no ghost values, no extrapolation into the solid, and no coupling between the velocity components. We give the step restriction the moving wall imposes, $\Delta t |\mathbf{u}_b| \lesssim \alpha_P V / A_{\text{wall}}$, and identify what it depends on.

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1 Scope

The stationary formulation is assumed known; it is described in `doc/freeslip_ibm_part2.pdf` and implemented in `SIMPLEC_faceArea_noSlip.ipynb`. We recall only what is needed.

The incompressible Navier–Stokes equations

$$\frac{\partial \mathbf{u}}{\partial t} + \nabla \cdot (\mathbf{u}\mathbf{u}) - \nabla \cdot (\nu \nabla \mathbf{u}) = -\nabla p + \mathbf{g}, \quad \nabla \cdot \mathbf{u} = 0, \quad (1)$$

are discretised by a collocated finite-volume method on a uniform, doubly periodic Cartesian mesh, and advanced by implicit Euler with a SIMPLEC pressure–velocity coupling in the arrangement used by OpenFOAM’s `simpleFoam`. A rigid body B is described by a level set φ , negative inside. Each face f carries the fraction $\theta_f \in [0, 1]$ of its area that lies in the fluid, and each cell P the fraction α_P of its volume. The fluid part of a cut cell is a polygon bounded by the partially open axis faces and by a wall segment, and the two are tied by the *closure identity*

$$\sum_f \theta_f \mathbf{n}_f A_f + \mathbf{n}_{\text{wall}} A_{\text{wall}} = \mathbf{0}. \quad (2)$$

We write $\mathbf{S}_w \equiv \sum_f \theta_f \mathbf{n}_f A_f = -\mathbf{n}_{\text{wall}} A_{\text{wall}}$ throughout; in the code its two components are `Swx`, `Swy`, and $A_{\text{wall}} = |\mathbf{S}_w|$. Crucially, $\mathbf{S}_w$ is not computed independently: it is *defined* by (2) from the same θ that the flux operators use, so (2) holds to roundoff by construction.

The body now moves, with prescribed position $\mathbf{x}_b(t)$ and velocity $\mathbf{u}_b(t) = \dot{\mathbf{x}}_b(t)$. All of θ , α , $\mathbf{S}_w$ and the fluid/solid mask become functions of time; the mesh does not move.

Remark 1. This is worth stating plainly because “moving body” usually triggers “ALE”. Here no axis face ever moves, so no face carries a mesh-velocity flux and no mesh-motion term appears in any operator. The entire modification lives on the wall segment.

2 The moving cut cell

2.1 What the wall sweeps

Let $\Omega_P(t) = \text{cell}_P \cap \{\varphi(\cdot, t) > 0\}$ be the fluid part of cell P . Its boundary consists of (i) the wet portions of the axis faces, which lie in fixed planes and therefore sweep no volume, and (ii) the wall segment, which moves with velocity $\mathbf{u}_b$. Reynolds transport gives immediately

$$\frac{d(\alpha_P V)}{dt} = \int_{\text{wall}} \mathbf{u}_b \cdot \mathbf{n}_{\text{wall}} dS \simeq (\mathbf{u}_b \cdot \mathbf{n}_{\text{wall}}) A_{\text{wall}} = -\mathbf{u}_b \cdot \mathbf{S}_w. \quad (3)$$

Note the sign: $\mathbf{n}_{\text{wall}}$ points out of the fluid, into the body, so a body advancing into the fluid has $\mathbf{u}_b \cdot \mathbf{n}_{\text{wall}} < 0$ and the cell loses volume, as it must.

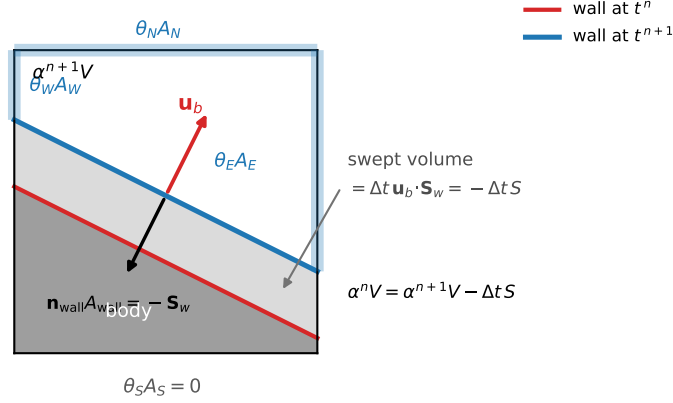


Figure 1: A cut cell over one step. The axis faces are fixed in space; only their wet fractions θ_f change. The shaded band is the volume the wall sweeps, which is $-\Delta t S_P = \Delta t \mathbf{u}_b \cdot \mathbf{S}_w$ and is the *only* quantity the motion contributes to the discretisation.

term	body at rest	body moving
continuity	$\sum_f \theta_f \phi_f = 0$	$\sum_f \theta_f \phi_f = \mathbf{u}_b \cdot \mathbf{S}_w$
convective	no wall term	no wall term
pressure	$p_P \mathbf{n}_{\text{wall}} A_{\text{wall}}$, absorbed by (2)	same
viscous, free slip	no wall term	no wall term
viscous, no slip	$\nu(0 - u_P) A_{\text{wall}}/d$	$\nu(u_b - u_P) A_{\text{wall}}/d$
unsteady	$\alpha_P V(u^{n+1} - u^n)/\Delta t$	$(\alpha^{n+1} u^{n+1} - \alpha^n u^n) V/\Delta t$

Table 1: Every term of the discretisation. Only the first and the last two rows differ, and the convective row is the one that is easy to get wrong.

Applying $\nabla \cdot \mathbf{u} = 0$ over the same region, with $\mathbf{u} \cdot \mathbf{n} = \mathbf{u}_b \cdot \mathbf{n}$ on the wall, gives the discrete continuity equation

$$\sum_f \theta_f \phi_f = -(\mathbf{u}_b \cdot \mathbf{n}_{\text{wall}}) A_{\text{wall}} = \mathbf{u}_b \cdot \mathbf{S}_w \equiv -S_P. \quad (4)$$

2.2 Continuity is the only equation that changes

Table 1 lists every term against its stationary counterpart.

The convective row deserves a word. The absolute mass flux through the wall is not zero, so one might expect a wall term $\mathbf{u}(\mathbf{u} \cdot \mathbf{n}) A_{\text{wall}}$. Writing the momentum balance over the *time-varying* region $\Omega_P(t)$ removes it: the flux that appears there is the relative one, $\mathbf{u}(\mathbf{u} - \mathbf{w}) \cdot \mathbf{n}$ with $\mathbf{w}$ the velocity of the boundary, and on the wall $\mathbf{w} = \mathbf{u}_b$ and $(\mathbf{u} - \mathbf{u}_b) \cdot \mathbf{n} = 0$. On the axis faces $\mathbf{w} = \mathbf{0}$ and the flux is the absolute one, exactly as in the stationary case. So the convective operator is untouched – provided the unsteady term is the one in Table 1, with the volume inside the difference.

2.3 Free slip needs nothing added

In the stationary method, no-penetration is not imposed anywhere: it *emerges*. For a uniform $\mathbf{c}$ in a cut cell, closure gives $\sum_f \theta_f \mathbf{c} \cdot \mathbf{n}_f A_f = \mathbf{c} \cdot \mathbf{S}_w$, and setting this to zero forces $\mathbf{c} \cdot \mathbf{n}_{\text{wall}} = 0$.

The same argument runs unchanged on a moving wall. Setting $\mathbf{c} \cdot \mathbf{S}_w = \mathbf{u}_b \cdot \mathbf{S}_w$, which is what (4) demands, forces

$$\mathbf{c} \cdot \mathbf{n}_{\text{wall}} = \mathbf{u}_b \cdot \mathbf{n}_{\text{wall}}. \quad (5)$$

The moving-wall no-penetration condition is a consequence of discrete continuity plus closure, and *the mass source is the boundary condition*. Nothing else is added for free slip: no ghost value, no extrapolation, no velocity coupling.

2.4 No slip adds one source

The wall traction of a no-slip wall moving at $\mathbf{u}_b$ is

$$\int_{\text{wall}} \nu \frac{\partial \mathbf{u}}{\partial n} dS \simeq \nu \frac{\mathbf{u}_b - \mathbf{u}_P}{d} A_{\text{wall}}, \quad d = \frac{\alpha_P V}{2A_{\text{wall}}}, \quad (6)$$

d being the mean thickness of the fluid the cell holds behind its wall segment, halved for a profile linear in the wall-normal direction. The $\mathbf{u}_P$ half is the diagonal term of the stationary scheme, unchanged: the body velocity does not enter it. The $\mathbf{u}_b$ half is a source. It vanishes for a wall at rest, which is why the stationary formulation needs no source at all.

Because $d \propto \alpha_P$, the coefficient $\nu A_{\text{wall}}/d$ grows like $1/\alpha_P$ as a cell is squeezed against the wall. The small-cell limit therefore *strengthens* diagonal dominance and drives $\mathbf{u}_P \rightarrow \mathbf{u}_b$, rather than degrading the system — the same property the stationary scheme has, now pulling the cell onto the moving wall instead of onto rest.

3 Which quantity is primary

Equations (3) and (4) are the same statement, so on paper one may take either as the definition and derive the other. Discretely the two choices are not equivalent, and only one of them works.

3.1 The two candidates

Candidate A: geometry first. Compute α^n and α^{n+1} from the level set and set

$$S_P = \frac{(\alpha_P^{n+1} - \alpha_P^n)V}{\Delta t}. \quad (7)$$

This is the textbook swept-volume prescription and it is what one writes down first.

Candidate B: flux first. Compute $\mathbf{S}_w$ from θ^{n+1} and set

$$S_P = -\mathbf{u}_b \cdot \mathbf{S}_w, \quad \text{then} \quad \alpha_P^n V := \alpha_P^{n+1} V - \Delta t S_P. \quad (8)$$

3.2 Why A fails

Candidate A presumes that α can be differentiated in time. It cannot, because α is not known exactly: `cellFraction` obtains it by sub-sampling the level set, with an error of order 10^{-3} per cell at 32×32 samples. Dividing that error by Δt amplifies it without bound. Table 2 compares the two candidates directly for a cylinder translating at $u_b = 0.37$ on an 81^2 mesh.

The last column is the more serious failure. The fluid region is a pure Neumann island for the pressure, so the pressure system is singular and needs a reference; following `fvMatrix::setReference` the reference is imposed by *doubling* one diagonal entry rather than by replacing a row, which keeps the matrix symmetric. As shown in §4.1, that construction returns the true solution *only* if the right-hand side is compatible. Candidate A is not compatible, so the entire mismatch is deposited in the reference cell as a spurious point mass source. Projecting the residual away restores solvability but leaves an $O(10^{-3})$ spurious dilatation, and the measured free-stream error of the resulting scheme is 50% of u_b .

Δt	displacement	$\max S_A - S_B / \max S_B $	$\sum_P S_{A,P}$
0.2000	0.599 cells	0.890	$+4.5 \times 10^{-4}$
0.1000	0.300 cells	0.199	-3.0×10^{-3}
0.0500	0.150 cells	0.280	-3.6×10^{-3}
0.0250	0.075 cells	0.561	-1.2×10^{-3}
0.0125	0.037 cells	0.166	$+3.1 \times 10^{-2}$

Table 2: Candidate A against candidate B. The difference does not converge as $\Delta t \rightarrow 0$ — it is quadrature noise divided by Δt — and $\sum_P S_{A,P}$, which must vanish, does not. For candidate B, $\sum_P S_{B,P}$ is 10^{-17} at every row.

3.3 Why B works

Candidate B never differentiates a sampled quantity. Two properties come free.

Proposition 1 (Exact compatibility). $\sum_P S_P = 0$ to roundoff, for every body position, every body velocity and every pattern of absorbed slivers.

Proof. $S_P = -\mathbf{u}_b \cdot \mathbf{S}_w$ with $\mathbf{S}_w = \sum_f \theta_f \mathbf{n}_f A_f$. Summing over all cells, each interior face appears twice with opposite normals and identical θ , so the sum telescopes; the periodic seam contributes nothing because θ there is computed once and copied. An absorbed sliver has all its faces closed, so $\mathbf{S}_w = \mathbf{0}$ and it contributes nothing at all. $\square$

Proposition 2 (Exactness for the divergence operator). For a uniform $\mathbf{u} \equiv \mathbf{u}_b$, $\text{div } \phi = \mathbf{u}_b \cdot \mathbf{S}_w = -S$ identically.

Proof. $\mathbf{S}_w$ is assembled from exactly the θ that `interpolatePhi` multiplies the face velocities by. The two expressions are the same sum. $\square$

The second half of (8), $\alpha^n V := \alpha^{n+1} V - \Delta t S$, is a backward Euler integration of the volume ODE (3). It is the classical resolution of the geometric conservation law — *take the volume from the flux, not the flux from the volume* — applied per step. Because α^n is recomputed from the current geometry every step and never accumulated, it cannot drift.

4 The discrete geometric conservation law

Proposition 3 (Free-stream preservation). Let the body translate rigidly at $\mathbf{u}_b$ and let $\mathbf{u}^n \equiv \mathbf{u}_b$ throughout the domain, with $p \equiv 0$. Then $\mathbf{u}^{n+1} \equiv \mathbf{u}_b$, $p \equiv 0$ is an exact solution of the discrete system, for any Δt , any body path, any mesh, and either wall condition.

Proof. Take the momentum row of a fluid cell P and apply the operator to the constant field $\mathbf{u}_b$. The row sum of each contribution is:

- unsteady: $\alpha^{n+1} V / \Delta t$;
- convective: the `fvm.divConvFlux` stencil has row sum $\phi_E - \phi_W + \phi_N - \phi_S = \text{div } \phi = -S$;
- viscous: the θ -weighted Laplacian has row sum zero, so it annihilates a constant, as it must;
- no-slip wall: diagonal $+\nu A_{\text{wall}}/d$, matched on the right-hand side by $+\nu A_{\text{wall}} \mathbf{u}_b/d$ from (6); they cancel at $\mathbf{u}_P = \mathbf{u}_b$;
- pressure: `fvc.grd` is written through (2) as $\sum_f \theta_f (p_f - p_P) \mathbf{n}_f A_f$ and vanishes for uniform p .

The equation therefore reduces to

$$\left(\frac{\alpha^{n+1}V}{\Delta t} - S\right) \mathbf{u}_b = \frac{\alpha^n V}{\Delta t} \mathbf{u}_b, \quad (9)$$

which is precisely the definition (8) of α^n . Continuity is satisfied because $\text{div } \phi = -S$ identically for the uniform field (Proposition 2), so the pressure equation has zero right-hand side and $p \equiv 0$ solves it. $\square$

Remark 2 (The effective unsteady diagonal). The *diagonal* contribution of the convective operator is $\frac{1}{2} \text{div } \phi$, so unsteady and convective terms contribute $\alpha^{n+1}V/\Delta t - \frac{1}{2}S = \frac{1}{2}(\alpha^{n+1} + \alpha^n)V/\Delta t$ to the diagonal: the *average* of the old and new fluid volume. A cell emerging from the body, with $\alpha^n = 0$, still carries half its new volume on the diagonal and cannot lose positivity for that reason.

4.1 Compatibility and the doubled reference diagonal

Let L be the pressure matrix restricted to the fluid cells, so that $\mathbf{1}^\top L = \mathbf{0}$ (a Laplacian assembled from face fluxes has columns summing to zero, and solid cells decouple because all their faces are closed). The reference is imposed as $L' = L + \mathbf{e}_r \mathbf{e}_r^\top d_r$ with $d_r = L_{rr}$.

Suppose $\mathbf{1}^\top \mathbf{b} = 0$ and let p solve $L'p = \mathbf{b}$. Then $Lp = \mathbf{b} - \mathbf{e}_r d_r p_r$, and left-multiplying by $\mathbf{1}^\top$ gives $d_r p_r = 0$, hence $p_r = 0$ and $Lp = \mathbf{b}$ exactly. The doubled diagonal is therefore *transparent* for a compatible right-hand side: it selects the solution that vanishes at the reference cell, and perturbs nothing.

For an incompatible right-hand side the same algebra gives $d_r p_r = \mathbf{1}^\top \mathbf{b} \neq 0$, and the whole defect appears as a point source at the reference cell. This is why exact compatibility, and not merely small incompatibility, is the property that matters — and why candidate B, which has it by construction, is preferable to candidate A plus a projection.

5 Algorithm

5.1 Where the source enters

The step is a PIMPLE-style loop: the geometry is advanced once, then held fixed while n_{outer} SIMPLEC iterations converge the nonlinearity.

1. Move the body to $t + \Delta t$ and rebuild θ , α , $\mathbf{S}_w$, A_{wall} , d and the fluid/solid mask.
2. Form $S = -\mathbf{u}_b \cdot \mathbf{S}_w$ and $\alpha^n V = \alpha^{n+1} V - \Delta t S$.
3. Assemble the step-invariant sources: $\alpha^n V \mathbf{u}^n / \Delta t$, the no-slip wall source $\nu A_{\text{wall}} \mathbf{u}_b / d$, and the blanking source that drives the solid interior to $\mathbf{u}_b$.
4. For each outer iteration: assemble the momentum operators, solve the predictor, form $\mathbf{u}_{HbyA}$ and ϕ_{HbyA} , and solve

$$\nabla \cdot (r_{A_t U} V \nabla p) = \nabla \cdot \phi_{HbyA} + S, \quad (10)$$

then correct $\phi = \phi_{HbyA} - r_{A_t U} V \nabla p$ so that $\nabla \cdot \phi = -S$.

Equation (10) is the entire algorithmic modification: one added term on the right-hand side of the pressure equation.

5.2 Constant flow rate

A doubly periodic box leaves the mean flow undetermined, so it must be pinned. The solver does so by superposition: the whole pass is affine in $(\text{source}, p^{\text{old}}, S)$, so one pass with $g_x = 0$ and one with $g_x = 1$ give the exact g_x that lands the bulk velocity on its target, with no gain to tune. The moving-wall contributions — S , the wall source, the blanking source — all belong to the $g_x = 0$ problem, so the superposition is untouched by them. Setting the target to zero is the natural choice for a body stirring an otherwise quiescent fluid, and is what makes the reported force meaningful.

5.3 Re-interpolating the flux across a geometry change

One line does not follow from the formulation but matters as much as any of it. The flux ϕ^n carried from the previous step lives on the faces of the *old* cut mesh: a face that has just opened has no history there, and one that has just closed carries a history that no longer exists. Before the outer loop starts, ϕ is therefore re-interpolated from $\mathbf{u}^n$ through the *new* θ .

Nothing is lost: this is only the initial guess for the convective operator, and the flux the step returns is the Rhie–Chow projected one either way. What is gained is that for a fluid translating rigidly with the body the guess lands *exactly* on the fixed point, so free-stream preservation is exact from the first outer iteration. Without it the fixed point is still exact but is only approached at roughly a factor of two per outer iteration, which costs two to four digits at practical n_{outer} — see Table 3.

6 Verification

6.1 Free-stream preservation

Fill the domain with $\mathbf{u} = \mathbf{u}_b$ and translate the body at that velocity. The fluid moves rigidly with the body: no relative motion, no pressure gradient, no traction, so $\mathbf{u} \equiv \mathbf{u}_b$ is the exact solution and any departure is a discretisation error. The test exercises the two-level unsteady term, the mass source, the moving-wall traction and the blanking simultaneously, and a sign error in any of them gives an $O(1)$ result.

n_{outer}	max $ \mathbf{u} - \mathbf{u}_b $ after 5 steps			
	1	2	3	5
$u_b = 0.37$, with §5.3	5.6e-16	5.6e-16	5.6e-16	6.1e-16
$u_b = 2.00$, with §5.3	3.6e-15	4.0e-15	3.6e-15	3.6e-15
$u_b = 0.37$, without §5.3	1.6e-03	1.4e-04	6.9e-05	2.4e-05
$u_b = 2.00$, without §5.3	1.2e-01	1.2e-02	6.5e-03	2.5e-03
$u_b = 0.37$, candidate A of §3	6.8e-02	6.3e-02	6.0e-02	5.6e-02
$u_b = 2.00$, candidate A of §3	1.8e+00	1.8e+00	1.9e+00	1.9e+00

Table 3: Free-stream preservation, $n_x = 81$, $\nu = 1$, $\Delta t = 0.025$, no slip. Machine precision at any body speed and from the first outer iteration. Candidate A does not converge with n_{outer} at all: its error is not a splitting error but a wrong fixed point.

6.2 Mass conservation

Over an oscillating-cylinder run, $\max_P |\text{div } \phi + S| \sim 10^{-16}$ and $|\sum_P (\text{div } \phi + S)| \sim 10^{-17}$ at every step, including steps in which slivers are absorbed or released. No projection is applied anywhere. See the bottom panel of Fig. 3.

6.3 Consistency of the swept volume

$\alpha^n V = \alpha^{n+1} V - \Delta t S$ is a backward Euler integration of (3), so it must converge to the geometric α at the old time. Table 4 confirms it. The comparison is restricted to cells that are cut in both configurations, since a cell crossing the sliver threshold in between changes by ALPHA_MIN for reasons unrelated to the time integration.

Δt	mean $ \alpha^n - \alpha_{\text{geom}} $	order	cells with $\alpha^n < 0$
0.4000	2.565e-01	—	17
0.2000	8.638e-02	1.6	12
0.1000	2.260e-02	1.9	2
0.0500	9.147e-03	1.3	0
0.0250	3.099e-03	1.6	0
0.0125	1.347e-03	1.2	0

Table 4: The swept volume against the geometry, $u_b = 0.37$, $n_x = 81$. The exactness of Table 3 is not bought by evolving a volume that has come loose from the body.

6.4 Galilean invariance

A fixed body in a bulk flow $+U$ and a body translating at $+U$ through a fluid of zero bulk velocity differ by a Galilean shift, so the streamwise forces must satisfy $F_x^{\text{mov}} = -F_x^{\text{fix}}$. The two are computed through entirely different terms — the moving case exercises the mass source, the wall source and the two-level unsteady term, none of which the fixed case touches — so the comparison is an independent check of the whole construction.

configuration	F_x	ripple
fixed body, bulk velocity $+U$	+5.66034	± 0.00000
body at $+U$, bulk velocity 0	-5.69040	± 0.15625
relative difference		0.53%

Table 5: Galilean invariance. $U = 0.5$, $\nu = 1$, $n_x = 81$, $\Delta t = 0.025$, no slip; each case run to $t = 30$ and averaged over the last 400 steps. The moving case carries a ripple of $\pm 2.8\%$ at the frequency U/h at which the body slides from one cut pattern to the next — the characteristic signature of a non-body-fitted cut-cell method, and the same effect that makes the cylinder drag a poor instrument for measuring the order of convergence. The means agree to well inside it.

6.5 Physical behaviour

Figures 2 and 3 show a cylinder oscillating along x in an otherwise quiescent periodic box, no slip, $\nu = 1$, $n_x = 81$. The impulsive start produces the expected added-mass spike; thereafter the force follows the body velocity with the phase lag of a viscous flow, and the bulk velocity stays at its target of zero to 10^{-18} .

7 Step restriction

The row sum of the unsteady plus convective operator is exactly $\alpha^n V / \Delta t$, so it changes sign when α^n does. From (8),

$$\alpha_P^n V = \alpha_P^{n+1} V + \Delta t \mathbf{u}_b \cdot \mathbf{S}_w \implies \boxed{\Delta t |\mathbf{u}_b| \lesssim \frac{\alpha_P V}{A_{\text{wall}}} = 2d} \quad (11)$$

The wall must not sweep through more than a wall-adjacent cell holds. Three things follow.

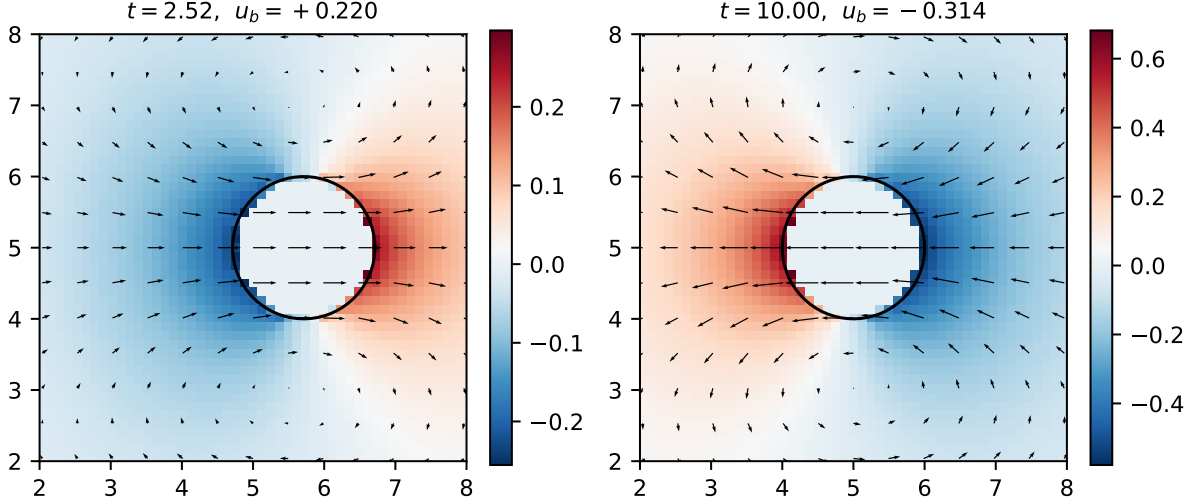


Figure 2: Pressure and velocity around the oscillating cylinder at two instants.

- It is a condition on the *smallest wet cell*, not on the mesh spacing. Refining the mesh does not tighten it in any simple way; ALPHA_MIN, which sets the smallest α admitted, does.
- It is not fatal when violated in a few cells. The diagonal there is still dominated by the viscous and wall terms, which grow like $1/\alpha_P$; what is lost is positivity of the row sum, not of the matrix. Runs with two or three violating cells out of 6380 are indistinguishable from clean ones. A large count means the step is genuinely too big.
- The solver reports $\min_P \alpha_P^n$ and the number of violating cells at every step.

8 Limitations

First order in time. Implicit Euler, and the volume integration in (8) inherits it. A second-order extension (BDF2) needs α at three levels and a three-level form of the argument in §4; the structure carries over but the bookkeeping does not, since α^{n-1} would then have to be stored rather than reconstructed.

First order in space for no slip. Unchanged from the stationary method: free slip is second order because it needs no wall-normal distance, no slip is first order because the estimate $d = \alpha_P V / (2A_{\text{wall}})$ is first-order accurate. Measured L_2 orders on a curved wall are 2.21, 2.33, 2.06, 2.14 (free slip) against 0.75, 1.27, 0.91, 0.90 (no slip).

Absorbed slivers lose their content. A cell crossing ALPHA_MIN is deactivated and hands over no momentum, an $O(\text{ALPHA_MIN} \cdot V)$ defect and the same one the stationary method has. Mass conservation is *not* affected: an absorbed cell has all faces closed, hence $\mathbf{S}_w = \mathbf{0}$, hence no source and no contribution to the compatibility sum. Cell merging rather than absorption is the clean fix and would also relax (11).

Prescribed motion. The hydrodynamic force is available — the pressure part is $-\sum_P p_P \mathbf{S}_w$ by (2), and the viscous part is the reaction to the traction already assembled in (6) — so closing a rigid-body ODE is short. A light body would need the added-mass sub-iteration inside the outer loop.

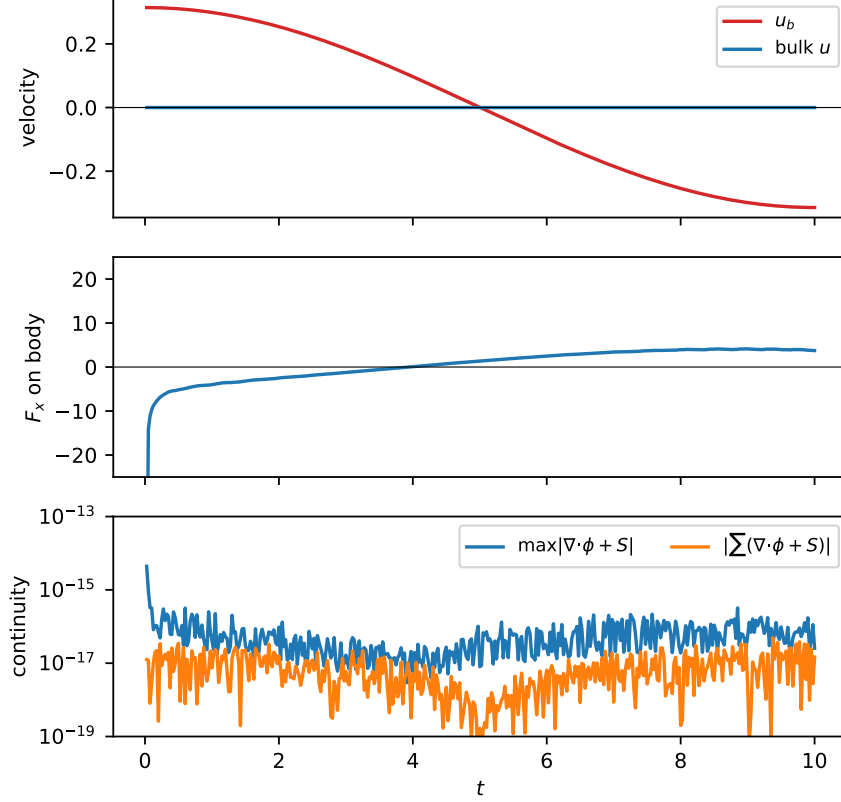


Figure 3: Oscillating cylinder: body and bulk velocity, streamwise force, and the two continuity diagnostics. The force axis is clipped; the impulsive start reaches -60 .

One fluid region. The pressure reference is a single cell, chosen as the fluid cell farthest from the body so that the body cannot swallow it. Should the body ever split the fluid in two, one reference per region would be required.

9 Summary

A rigid body moving through a fixed cut-cell mesh changes exactly one equation. Continuity acquires the source $S = -\mathbf{u}_b \cdot \mathbf{S}_w$, and that source is simultaneously the moving-wall no-penetration condition, so free slip needs nothing else and no slip needs only the $\mathbf{u}_b$ half of its wall traction.

The choice that decides whether the scheme works is which of the volume change and the wall flux is primary. Taking the flux as primary and the volume change from it gives exact compatibility of the pressure system, exact consistency with the divergence operator, and a discrete geometric conservation law that holds by construction; the reverse ordering differentiates a sampled volume and fails outright. With one further line — re-interpolating the carried flux onto the new geometry — a free stream is preserved to machine precision at any body speed and from the first outer iteration, and mass is conserved to roundoff without any projection.